

- **Products and services:** Lithium-ion battery (LFP and NMC, design services and consulting for battery manufacturing)
- **Battery capacities developed:** 2 to 10kW, fixed and swappable, for B2B segment
- **Category:** Two-wheelers, three-wheelers, and energy storage solutions
- **Geography:** Currently operating in India and in discussions with international clients
- **Production stage:** The design and prototype have received approval, and production is likely to commence this month
- **Manufacturing plant:** Mysore funding/investment: raised US\$ 1.2m in seed funding

Notably, most electric two-wheelers being retailed in India today come equipped with chargers that consume 4kWh of power. Sinha claims that the technology used in EMO Energy's batteries allows them to charge at a 4kWh rate via a 2kWh line.

He explains, "Motors inside our battery packs help the battery charge faster. These also keep pushing the hot liquid, taking care of the extra heat generated while charging."

In a nutshell, the startup claims that its 2kW batteries can be charged for about 2km every minute, translating to around a 40 to 45km range in 20 minutes.

Sinha adds, "Additionally, we have strong partnerships with fleet and swap operators. We are also actively collaborating with charge point operators to expand the availability of AC001 sockets across the country."

Having raised \$1.2 million in seed funding so far, the startup says it has also developed 5kWh and 10kWh batteries but will not be putting the same into production for now.

"We will only be manufacturing 2kWh batteries as we believe these are a perfect fit for E2Ws and E2Ws, as these two sectors are leading the electric vehicle adoption in the country," he says.

The startup says it is utilising a digital Industry 4.0 setup, which includes robots and an interconnected assembly line to ensure the production of safe batteries. Sinha adds that professionals at EMO Energy have a cumulative experience of 45 years in the battery industry and have previously worked with companies such as Tesla, Rivian, Ola, Sun, Ather, and Log9.

Its collaborations extend to inverter and energy storage companies



Mrinal Sinha, Co-Founder

to provide them with batteries for a second life. Sinha claims that EMO Energy's batteries can be used for almost 10 years.

At the end of the cycle, the disassembled cells will be supplied to various existing extractors for conversion into a black mass. "The materials extracted from this process will then be provided to the cell companies we work with, enabling them to build cells for our battery packs," Mrinal concludes.

—Ashish Chatterjee

3 ALTIGREEN TO BE THE FIRST TO INTRODUCE PASSENGER ELECTRIC THREE-WHEELERS IN SEMI-URBAN AREAS

Altigreen, a Reliance-backed company with a decade of experience, will debut in the passenger electric three-wheeler segment next month and is considering entering the commercial electric four-wheeler market in FY25

Originally, Altigreen was conceived as an organisation that would design, develop, manufacture, and supply specific electric vehicle components to other

original equipment manufacturers (OEMs). Debashish Mitra, Director of Sales & Marketing at Altigreen, explained in a conversation with Electronics For You that the startup

had sent out specific components and drivetrains to electric two-wheeler, three-wheeler, and four-wheeler OEMs.

"The growth in that segment